

Electrifying industrial hydrogen peroxide production via soft interfacial molecular mediation

Received: 27 August 2024

Accepted: 7 August 2025

Published online: 22 September 2025

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Hydrogen peroxide is manufactured industrially via the anthraquinone autoxidation process—a typical thermocatalytic non-aqueous method. Despite a high interest in using renewable electricity to drive such processes, electrifying non-aqueous syntheses remains a substantial challenge. Here we present a multi-phase electrochemical anthraquinone autoxidation process that leverages an aqueous–non-aqueous interfacial proton-coupled electron transfer method facilitated by heterogeneous molecular mediation. This design enables the reduction of aqueous anthraquinones with high efficiency at high current densities, using only carbon electrodes. The method operates with high selectivity through a quinhydrone intermediate and prevents the over-reduction of aromatics during thermocatalytic hydrogenation. This approach combines the benefits of aqueous electrochemistry with those of the traditional non-aqueous process to achieve high current density electrochemistry with rapid kinetics and mass transport, while avoiding unwanted electrolyte in the hydrogen peroxide product. This strategy bridges aqueous electrochemistry with non-aqueous chemistry and establishes a framework for the electrification and decentralization of other non-aqueous chemical processes.

Many industrial chemical manufacturing processes, including hydrogen peroxide (H_2O_2) production, are carried out in non-aqueous solvents due to considerations of solubility, reactivity and separation¹. For instance, reaction between organic molecules and hydrogen in organic solvents using heterogeneous catalysts allows high reactant concentrations and facilitates catalyst–product separation (Fig. 1a). In contrast with conventional thermochemical methods, electrochemical driving forces and kinetics are tunable via the electrode potential^{2,3}, enabling selective reactions without the need for stoichiometric reagents⁴. Performing electrochemistry in low-polarity, low-conductivity, non-aqueous systems, however, poses major challenges. Introducing organic supporting electrolytes, such as tetrabutylammonium perchlorate or lithium bis(trifluoromethanesulfonyl)imide, can improve ionic

conductivity, but these salts risk contaminating the product and are incompatible with aqueous counter reactions (for example, water electrolysis)⁵. In such cases, charge balancing becomes difficult, as these large non-aqueous ions cannot serve as efficient proton or hydroxide carriers, often leading to ion accumulation and cell instability. Moreover, conventional ion-exchange membranes are not optimized for transport of bulky non-aqueous ions (Fig. 1b)⁶.

Aqueous electrochemical systems, using increasingly affordable renewable electricity, benefit from well-developed electrodes, membranes and low resistance^{7,8}. Yet aqueous solvents inherently limit the solubility of many organic substrates, and water-soluble products often become mixed with the electrolyte, complicating downstream separation⁹ (Fig. 1c).

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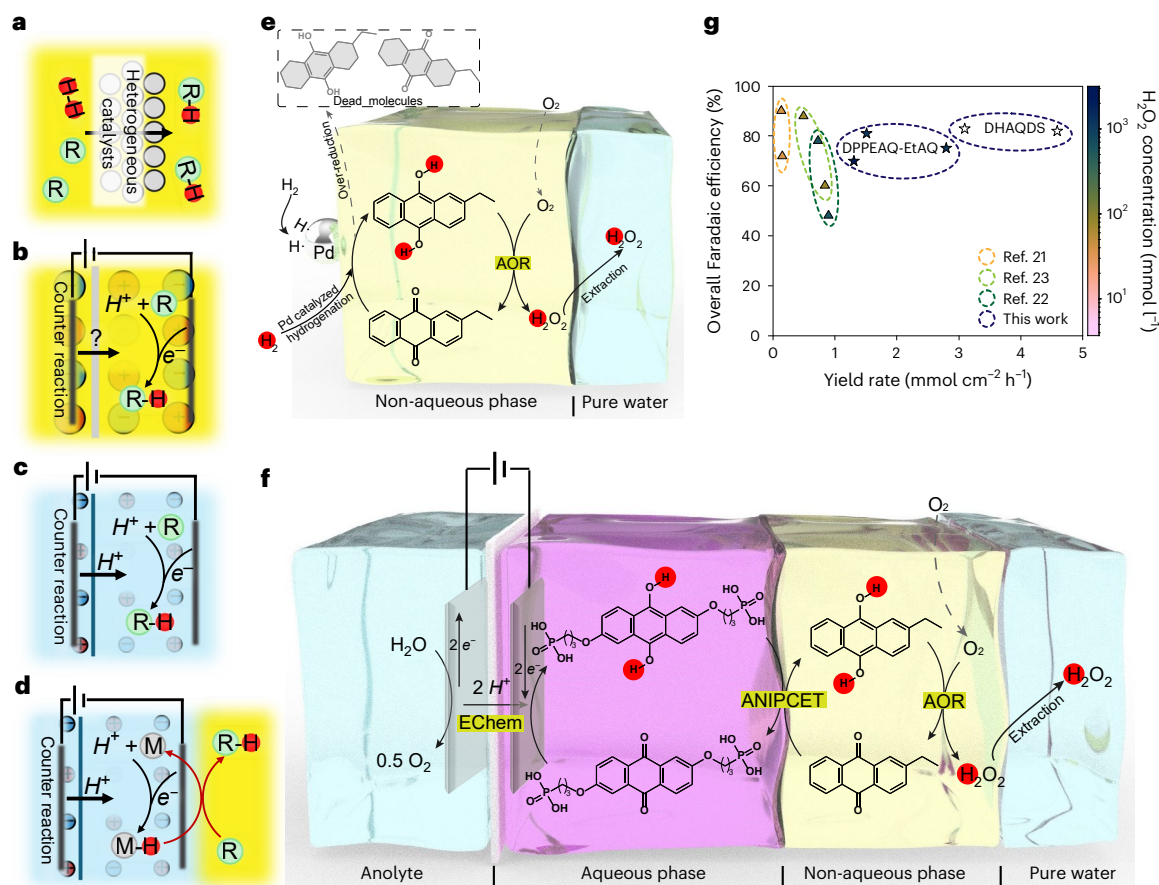


Fig. 1 | Electrifying non-aqueous synthesis with aqueous electrochemistry with H_2O_2 production as a demonstration. **a**, Non-aqueous synthesis using heterogeneous catalysts. Using hydrogenation as an example, reactants (R) form products (R-H) that can be separated physically. **b**, Non-aqueous electrocatalysis. Although R can be electrochemically reduced into R-H, counter reactions, charge carriers and mixing of electrolytes (background charged spheres) can be problematic. **c**, Aqueous electrocatalysis. Although the aqueous environment allows the implementation of developed membranes, features low resistance and balanced charge carriers during electrocatalysis, separation and concentration of the aqueous-soluble products can be problematic. **d**, Aqueous electrocatalysis mediated non-aqueous chemistry. ANIP CET (red arrows) mediated by aqueous redox couple (M/M-H) allows electron-proton pairs traversing the soft interface, prohibiting product contamination by transport of supporting electrolytes into the non-aqueous phase. **e**, Schematic of industrial H_2O_2 production by the conventional AO process. **f**, Electrochemical H_2O_2 production by the e-AO process. **g**, Comparison of different strategies for electrifying the AO process^{21–23}. Colours of the points indicate the accessible or demonstrated concentrations of produced H_2O_2 , which are reported in Supplementary Table 1. The coloured shading represents the trend of the properties with regard to different strategies.

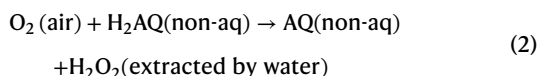
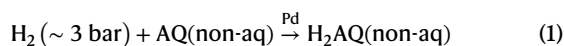
We introduce an approach that leverages aqueous electrochemistry to drive non-aqueous reactions^{10,11} through aqueous–non-aqueous interfacial proton-coupled electron transfer (ANIP CET). This method retains the advantages of aqueous electrochemical infrastructure, efficient charge transport, membrane compatibility and sustainable operation, while enabling selective chemistry in a non-aqueous phase, without introducing extraneous electrolytes into the product stream (Fig. 1d). The soft interface created between aqueous and non-aqueous phases provides unique opportunities for the optimization of mass transport for interfacial reactions.

Hydrogen peroxide, a versatile and environmentally benign oxidant, serves as an ideal case study to demonstrate this concept. Electrochemical synthesis of H_2O_2 in aqueous phases is doable¹², but the product can be mixed with impurities, such as ions and additives in the electrolytes¹³ (Fig. 1c), necessitating additional costly purification¹⁴. Although great effort has been made in the development of direct electrolysis methods for pure H_2O_2 solution production, including utilizing gas diffusion electrode (GDE) engineering¹⁵, metal hydrides¹⁶ and solid electrolytes in multi-chamber cells^{17–19}, substantial challenges remain in achieving high H_2O_2 concentrations without a notable decline in Faradaic efficiency (FE). This occurs because increasing the H_2O_2 concentration shifts the equilibrium of the oxygen reduction reaction (ORR), suppressing the

2e^- pathway to H_2O_2 and promoting the competing 4e^- pathway towards water¹⁷. Additionally, the direct synthesis of H_2O_2 through the 2e^- ORR relies on GDEs to facilitate O_2 mass transport, whereas GDEs are highly susceptible to performance failures caused by flooding^{3,20}.

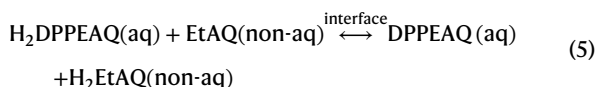
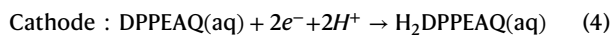
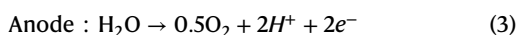
Until now, essentially all industrial H_2O_2 is still manufactured through the thermocatalytic anthraquinone autoxidation (AO) process¹. The AO process is one of the most successful non-aqueous solution-phase chemical engineering processes, highly effectively producing concentrated H_2O_2 with straightforward separation from organic solvents under relatively mild operating conditions and costing little energy. This continuous process involves three key steps (Fig. 1e): (1) thermocatalytic hydrogenation of non-aqueous-soluble anthraquinones (AQs) by hydrogen gas (equation (1)), catalysed by palladium (Pd); (2) anthraquinone oxidation reaction (AOR), effected by introducing air into the solution containing the resulting reduced AQs, quantitatively producing H_2O_2 (equation (2)); and (3) extraction of H_2O_2 by water. Because the non-aqueous solution and the AQs utilized are not soluble in water, whereas H_2O_2 has a low solubility in non-aqueous solutions, the AO process is capable of producing high-concentration H_2O_2 solution with high selectivity. The hydrogenation process to form the H_2AQ , however, requires an external hydrogen input and palladium-based hydrogenation catalysts. Additionally, a small fraction

of the AQs can be over-reduced into unfunctional compounds (for example, octahydroanthraquinones)¹.



Although researchers have tried phase-transfer catalysis²¹, emulsions²² and Pd membrane devices²³ to electrify the AO process and overcome the electrochemical limitations of non-aqueous solvents used for AOR, these reported results can hardly facilitate highly selective anthraquinone hydrogenation under high current densities due to mass transport limitations from anthraquinones²¹ or hydrogen atoms²³.

Here we report an electrochemical hydrogenation–anthraquinone autooxidation (abbreviated as e-AO) process with four separate steps (Fig. 1f and Supplementary Fig. 1): (1) electrochemical aqueous anthraquinone hydrogenation using a carbon electrode; (2) ANIPCET; (3) non-aqueous AOR; and (4) H₂O₂ extraction. Different redox-active AQs were used as hydrogen-atom carriers in the two immiscible phases. Owing to great kinetics and the selectivity of electrochemistry of aqueous AQs and ANIPCET, the e-AO process is capable of producing H₂O₂ with high concentration, high overall Faradaic efficiency and a high H₂O₂ yield rate^{21–23} (Fig. 1g). As a demonstration, we utilized 2,6-bis(3-phosphonopropyl-1-oxy)anthraquinone (DPPEAQ, aqueous AQ)²⁴ in the neutral pH aqueous phase (aq), and 2-ethylanthraquinone (EtAQ, non-aqueous AQ) in the non-aqueous phase (non-aq). Each AQ displays high solubility in its corresponding phase and poor solubility in the other, ensuring exceedingly low AQ loss caused by crossover into the other phase. During the operation, DPPEAQ is first reduced in an aqueous electrochemical cell (equations (3) and (4)). Then, H₂DPPEAQ serves as a mediator, reducing EtAQ through ANIPCET (equation (5)), a heterogeneous process.



Subsequently, the AOR and H₂O₂ extraction steps are conducted in the same manner as in the AO process.

This e-AO process inherits the benefits from the AO process while circumventing the consumption of H₂ and further improving the hydrogenation selectivity in a metal-free half-cell. The aqueous AQ and non-aqueous AQ used could function as electrical capacity buffers, further facilitating decentralized H₂O₂ production powered by intermittent renewable electricity. Moreover, using aqueous redox-active molecules in a flow cell to mediate the 2e[−] ORR (in a chemical reaction tank outside the electrochemical cell) eliminates the need for GDEs, allowing utilization of robust hydrophilic carbon paper electrodes. The ANIPCET process at the soft interface bridges the mass transport and thermodynamics of the aqueous and non-aqueous phases, transcending the challenges associated with electrochemical hydrogenation in the non-aqueous phase and improving the selectivity of thermocatalytic hydrogenation during the conventional AO process. The non-aqueous phase also prevents the contamination of H₂O₂ by the solutes in the aqueous electrolytes. This multi-phase system demonstrates wide tunability of aqueous and non-aqueous redox molecules and organic solvents, providing ample space for further adaptation and insights to electrify other non-aqueous synthetic applications, from hydrogenation to other ion-coupled reduction or oxidation reactions²⁵.

Results and discussion

ANIPCET

The ANIPCET process is a reversible, non-Faradaic proton-coupled electron transfer (PCET) process occurring at the aqueous–non-aqueous phase boundary, requiring a PCET donor and acceptor in each phase. For aqueous electrochemistry and ANIPCET donors, we used aqueous-soluble AQs, which are developed for redox flow batteries, showing high reversibility and great efficiency under high current densities⁸. Non-aqueous AQs, as ANIPCET acceptors, are utilized for producing H₂O₂ without being extracted into the aqueous H₂O₂ product.

The aqueous and non-aqueous AQs must remain within their respective phases exclusively to yield a pure aqueous solution of H₂O₂.

For the solvent serving as the non-aqueous phase, a mixture of toluene (Tol) and 1-decanol (Dec) was used. Typical solvents (for example, a mixture of mesitylene and tributyl phosphate (TBP)) used in the AO process²⁶ are recommended for e-AO for improved solubility, selectivity and safety (Supplementary Table 2). Utilizing ultraviolet–visible spectrophotometry (UV–Vis), we ascertained that DPPEAQ is virtually insoluble in Tol/Dec and that EtAQ does not dissolve in water (Supplementary Figs. 2–4 and Supplementary Table 3), prohibiting phase crossover of AQs. Consequently, redox processes between H₂DPPEAQ and EtAQ are confined to the interface between the aqueous and the non-aqueous phase. During ANIPCET, the formation of semiquinone in each phase can further complicate the actual reaction mechanism (Supplementary Fig. 5). Yet, density functional theory (DFT) calculations indicate that the key PCET step occurs through a quinhydrone-like intermediate formed on the aqueous–non-aqueous interface^{27,28}. According to the computations, PCET is associated with a lower energy transition state than two-step electron and proton transfers (Supplementary Figs. 6 and 7)²⁹. Free energies of reaction products and intermediates can also be influenced by the solvation energies of different compositions of the non-aqueous solvents (Supplementary Fig. 8). We further experimentally examined the kinetics and thermodynamics of the ANIPCET process.

To investigate the impact of non-aqueous solvent composition on ANIPCET, kinetic experiments were conducted using water phase and Tol/Dec or Tol/TBP solvent systems with pumping in each phase, while allowing a fixed contacting area with the aqueous phase (Supplementary Figs. 9 and 10). The abbreviated label *n*Tol/*m*Dec denotes *n* ml toluene mixed with *m* ml 1-decanol; similarly for *n*Tol/*m*TBP. The rate of the ANIPCET reaction can be described by equation (6). Here *r* represents the normalized reaction rate at the interface; *t* is time, *V* is the volume of DPPEAQ solution and *A* is the area of the aqueous–non-aqueous interface. Figure 2a shows that, with the same solution in both phases, increasing the flow rates of both phases by 50% does not apparently change the *r* of the ANIPCET reactions. This indicates that the bulk mass transport for each phase does not limit the reaction rate. ANIPCET under the applied flow rate is reaction-limited. Figure 2b shows a near-proportional change of initial reaction rate (*r*_{initial}, proportional to the slope) at the beginning of ANIPCET, by varying the concentration of reactants in the respective solvents, showing apparent second-order kinetics (*r*_{initial} ≈ *k*₂[H₂DPPEAQ][EtAQ], where *k*₂ is the apparent rate constant at the beginning of ANIPCET) (Supplementary Table 4). Tol/TBP as the non-aqueous solvent increases the kinetics of ANIPCET, presumably due to the stronger solvation energy TBP provides for the quinhydrone intermediate and reduced EtAQ.

ANIPCET, as an interfacial reaction, has a reaction rate that is proportional to the reaction surface area. For practical applications, vigorous agitation can substantially increase the interfacial area between the aqueous and non-aqueous phases, resulting in the attainment of equilibrium within only 3 s (Fig. 2c and Supplementary Fig. 11). The state of charge (SOC) of the aqueous phase is defined as the molarity ratio of the fully reduced form of the molecules to the total molarity, and the changes of SOC were measured during tests to characterize

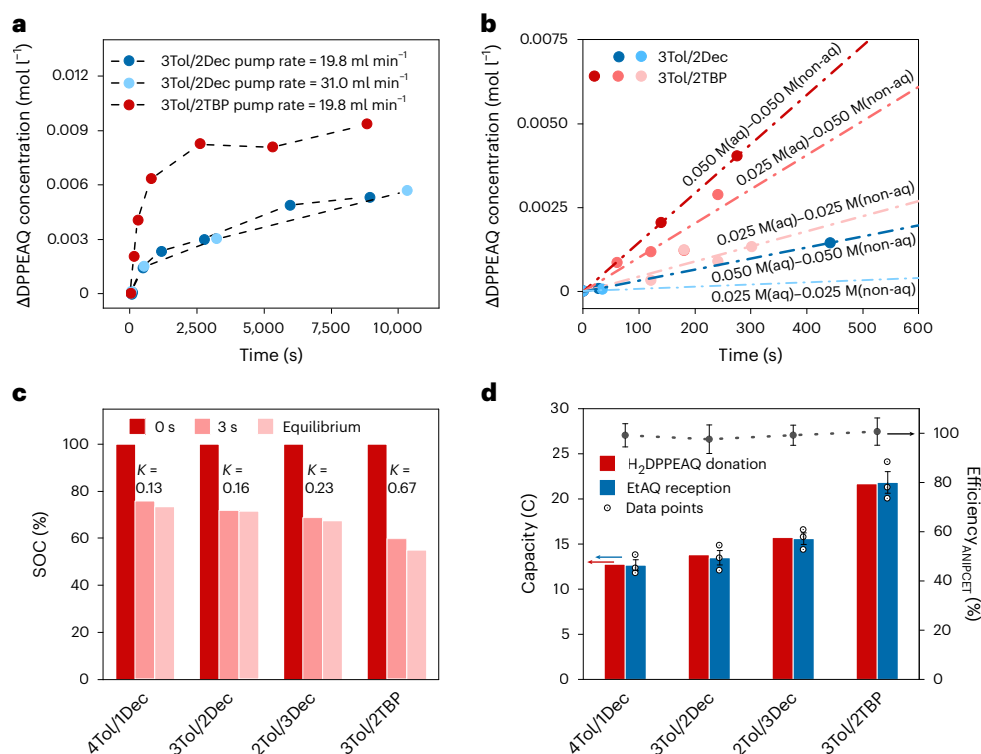


Fig. 2 | Kinetics and thermodynamics of ANIPCET. **a**, The temporal variation of the concentration of produced DPPEAQ during the process described by equation (5). The test was done with identical initial concentrations (0.050 mol l⁻¹) of the reactants. Non-aqueous solvents and pump rates were varied accordingly, as labelled. **b**, The temporal variation of the concentration of produced DPPEAQ at the beginning of the kinetic test. As labelled, the test was done with an initial solution containing 15 ml x mol l⁻¹ H₂DPPEAQ and 15 ml y mol l⁻¹ EtAQ (labelled as xM(aq) – yM(non-aq)) using various non-aqueous solvents. The slopes of the curves at the beginning (Supplementary Table 4)

are proportional to the initial reaction rates. **c**, SOC of H₂DPPEAQ/DPPEAQ solution during ANIPCET under vigorous stirring using non-aqueous solvents with different solvation energies for (H₂)EtAQ. *K* near equilibrium is labelled accordingly. **d**, Amount of proton–electron pairs transferring out of the aqueous phase and entering the non-aqueous phase. Number of electrons received by the non-aqueous phase is presented as mean value ± s.e.m., with individual data points shown. Efficiency_{ANIPCET} using various non-aqueous solvents was calculated accordingly.

the tendency of ANIPCET. The reaction quotients near equilibrium (*K*) under various non-aqueous solvents, defined by equation (7), are labelled in Fig. 2c. A stronger solvation of the product (H₂EtAQ) results in a larger *K*.

$$r = \frac{Vd[\text{DPPEAQ}]}{Adt} \quad (6)$$

$$K = \frac{[\text{DPPEAQ}]_{\text{eq}}[\text{H}_2\text{EtAQ}]_{\text{eq}}}{[\text{EtAQ}]_{\text{eq}}[\text{H}_2\text{DPPEAQ}]_{\text{eq}}} \quad (7)$$

After sufficient agitation, we measured the capacity of proton–electron pairs leaving the aqueous phase (DPPEAQ donation) and the corresponding capacity of proton–electron pairs entering the non-aqueous phase (EtAQ acceptance). Efficiency_{ANIPCET}, defined as the ratio of these capacities, was determined to be approximately 100% (Fig. 2d). A trade-off exists with respect to the choice of the solvent: EtAQ prefers a lower polarity solvent, whereas H₂EtAQ prefers higher polarity. Mixed solvents, as used in the industrial AO process (Supplementary Table 2), are recommended for optimized (H₂)EtAQ solubility, which is crucial for mass transport of the non-aqueous anthraquinones and concentration of the final H₂O₂ product.

H₂O₂ production using the e-AO process

Continuous H₂O₂ production through the e-AO process was conducted in a tandem reactor, with three major components: (1) aqueous electrochemistry (equation (4), labelled as EChem, defining efficiency_{EChem} as

proton–electron pairs DPPEAQ received divided by charges injected through the electrode) for aqueous AQ hydrogenation; (2) ANIPCET; and (3) AOR with water extraction (equation (2), defining efficiency_{AOR} as produced H₂O₂ in water divided by consumed H₂EtAQ). Overall Faradaic efficiency of the whole process, defined as the amount of H₂O₂ product divided by charges injected through the electrode, equals the product of the efficiencies of all steps. During ANIPCET and AOR, which require interfacial reaction/extraction between the aqueous and non-aqueous phase, mixers and separators were used to facilitate the phase mixing and separation, as labelled in Fig. 3a and Supplementary Fig. 12.

To quantify efficiency_{EChem} and stability of DPPEAQ, we first conducted a long-term charging–discharging cycling test of 6.5 ml of 0.1 mol l⁻¹ DPPEAQ in 0.33 mol l⁻¹ phosphate buffer (pH 7.3). The cell was cycled against ferro/ferricyanide dissolved in the same buffer solution in a nitrogen-filled glovebox. The cell was cycled for six days, retaining its pH at a value of 7.3 after cycling, and demonstrated a Coulombic efficiency (discharge capacity divided by charge capacity) of approximately 99%, with a capacity fade rate of approximately 0.2% per day (Fig. 3b). The high Coulombic efficiency during the cell cycling reflects the high efficiency_{EChem}. The fading was due to anthrone and dimer formation of DPPEAQ under near-neutral conditions³⁰, as determined by liquid chromatography–mass spectrometry (LC–MS) analysis (Supplementary Fig. 13). These side products could be recovered through electro-oxidation⁷ or chemical treatments³¹.

To evaluate the compatibility of ANIPCET with EChem, after a few days cycling of only DPPEAQ in aqueous buffer, we added 3 ml of 0.1 mol l⁻¹ EtAQ in 3Tol/2Dec (theoretical capacity: 57.9C) on top

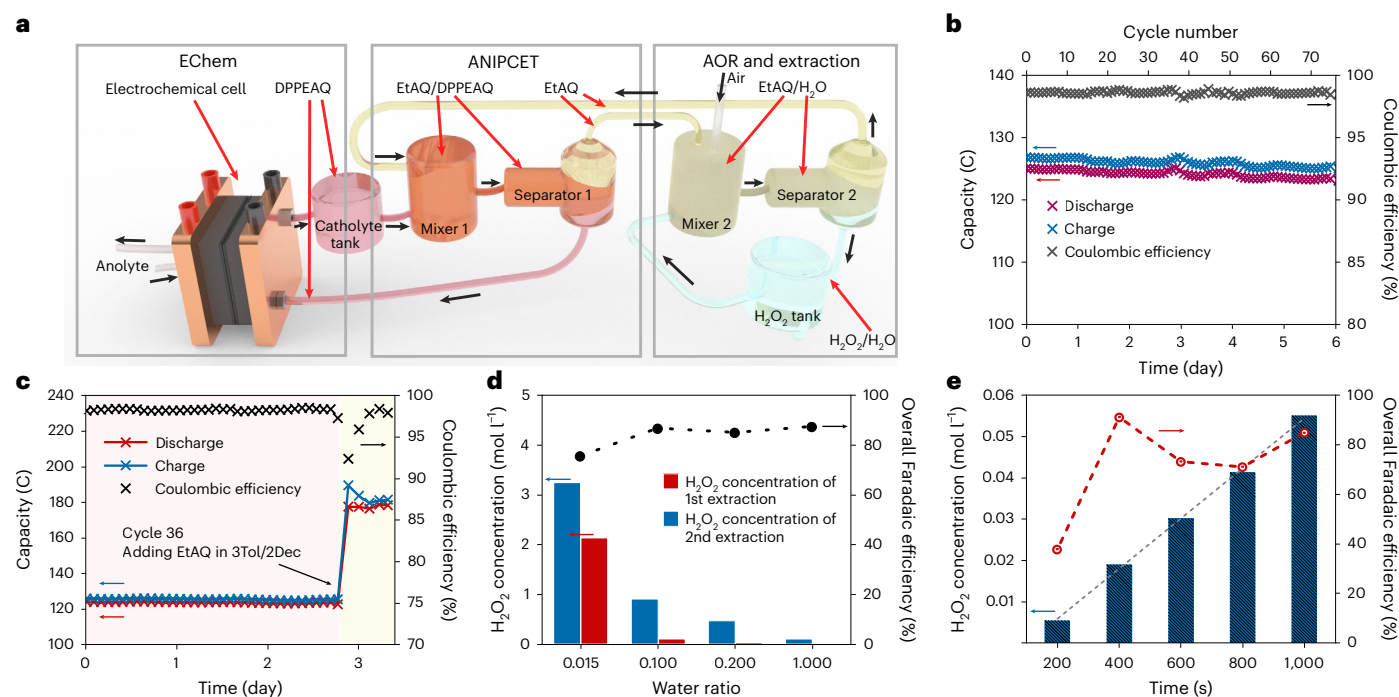


Fig. 3 | Performance of the e-AO process for H_2O_2 production. **a**, Device for H_2O_2 production through the e-AO process. The three major parts are framed. Components and materials are labelled accordingly. Black arrows indicate the flow direction during steady-state production. **b**, The charge and discharge capacity and Coulombic efficiency of the EChem process. **c**, The charge and discharge capacity and Coulombic efficiency of the EChem + ANIP CET process.

Cycling of DPPEAQ is shaded in red. Cycling of DPPEAQ with EtAQ is shaded in yellow. Non-aqueous EtAQ solution was added at the time labelled. **d**, The overall FE of the batchwise e-AO process and corresponding H_2O_2 concentration under various volumetric ratios of water to non-aqueous phase during the extraction of H_2O_2 . **e**, The temporal variation of overall FE and H_2O_2 concentration in the product during the continuous H_2O_2 production.

of the DPPEAQ solution during cell cycling. The cell cycling capacity was found to increase from approximately 125C to 180C (Fig. 3c), indicating that the non-aqueous EtAQ could be cycled along with the aqueous phase, presumably due to reversible ANIP CET and the similarity of redox potentials between the EtAQ and DPPEAQ. The initial drop of Coulombic efficiency in the first cycle was due to the presence of residual oxygen dissolved in the non-aqueous phase. A roughly constant Coulombic efficiency attained afterwards indicates that ANIP CET was successfully incorporated into the EChem process (charging and discharging) with high efficiency, indicating a high value of the product efficiency $\eta_{\text{EChem}} \times \text{efficiency}_{\text{ANIP CET}}$. Post-cycling LC-MS analysis provided evidence of the high selectivity of ANIP CET, showing only trace amounts of anthrone/dimer formation from EtAQ (Supplementary Fig. 14). The pH of the DPPEAQ solution is crucial because a pH lower than 7 leads to DPPEAQ precipitation, whereas a pH higher than 10 causes deprotonation of H_2EtAQ , forming a water-soluble anion, EtAQ^{2-} (Supplementary Fig. 15)²⁴.

Having confirmed the high efficiency of EChem and ANIP CET working in tandem, we next sought to include AOR in a batch-to-batch mode for hydrogen peroxide synthesis. First, we used DPPEAQ to mediate the reduction of 50 ml of 0.5 mol l^{-1} EtAQ in 3Tol/2Dec, which was added to the initially fully charged non-aqueous phase, and continued to charge the system until approximately 20% of EtAQ was reduced as H_2EtAQ . After that, the non-aqueous phase was separated and agitated with air to completely oxidize H_2EtAQ . We extracted the produced hydrogen peroxide twice for various volume ratios between water and the non-aqueous phase while we fixed the volume, concentration and SOC of the non-aqueous phase. The concentration of H_2O_2 products was determined by titration with KMnO_4 , from which the overall FE was calculated. The extraction efficiency was not constant for the different extraction volumes, due to the identical partition factor of H_2O_2 in aqueous/non-aqueous phases. As shown in Fig. 3d, with a water ratio of 0.1–1, after the second extraction, there is almost no H_2O_2 remaining in

the non-aqueous phase, demonstrating an overall FE of approximately 85%. The highest concentration of H_2O_2 produced in a single extraction trial was 3.2 mol l^{-1} (>10 wt%) at a water-to-non-aqueous solvent volumetric ratio of 0.015, although the first extraction did not go to completion: more than 30% of the H_2O_2 remained in the non-aqueous phase³². The overall FE is about 80–85% when extraction is sufficient, as verified by additional extraction.

Finally, we performed continuous H_2O_2 production in the tandem reactor under ambient conditions. During the electrolysis at 50 mA cm^{-2} in a 5-cm^2 flow cell, the cathode voltage remained stable (Supplementary Figs. 16 and 17). The H_2O_2 solution in the product tank was sampled intermittently and titrated by KMnO_4 to determine the concentration of product and FE. As expected, low FE was observed at the beginning due to capacity being injected into aqueous and non-aqueous AOs to build up the state of charge before reaching steady state. Subsequently, the steady state, overall FE of H_2O_2 production was about 80%, limited by efficiency_{AOR}, which can be improved by using Tol/TBP as non-aqueous solvents (Supplementary Fig. 18). It is also worth noting that the presence of residual oxygen in the non-aqueous phase after Separator 2 enters Mixer 1 can negatively influence the FE by oxidizing H_2DPPEAQ without forming H_2O_2 (Supplementary Figs. 19 and 20). During continuous production, the concentration of H_2O_2 increased almost linearly with time, reaching 0.055 mol l^{-1} (1.9 wt%) after 1,000 s (Fig. 3e). Operating the system at a higher current would require a larger AOR tank or higher oxygen pressure to accelerate the rate-limiting AOR in the demonstrated tandem reactor (Supplementary Fig. 21).

Generalization of the e-AO process

The e-AO concept allows for flexible selection of anode reactions, as well as choice of aqueous AOs and non-aqueous quinones, providing space for further optimization and other applications. In principle, the anodic reaction of e-AO could be any oxidation reaction, as long

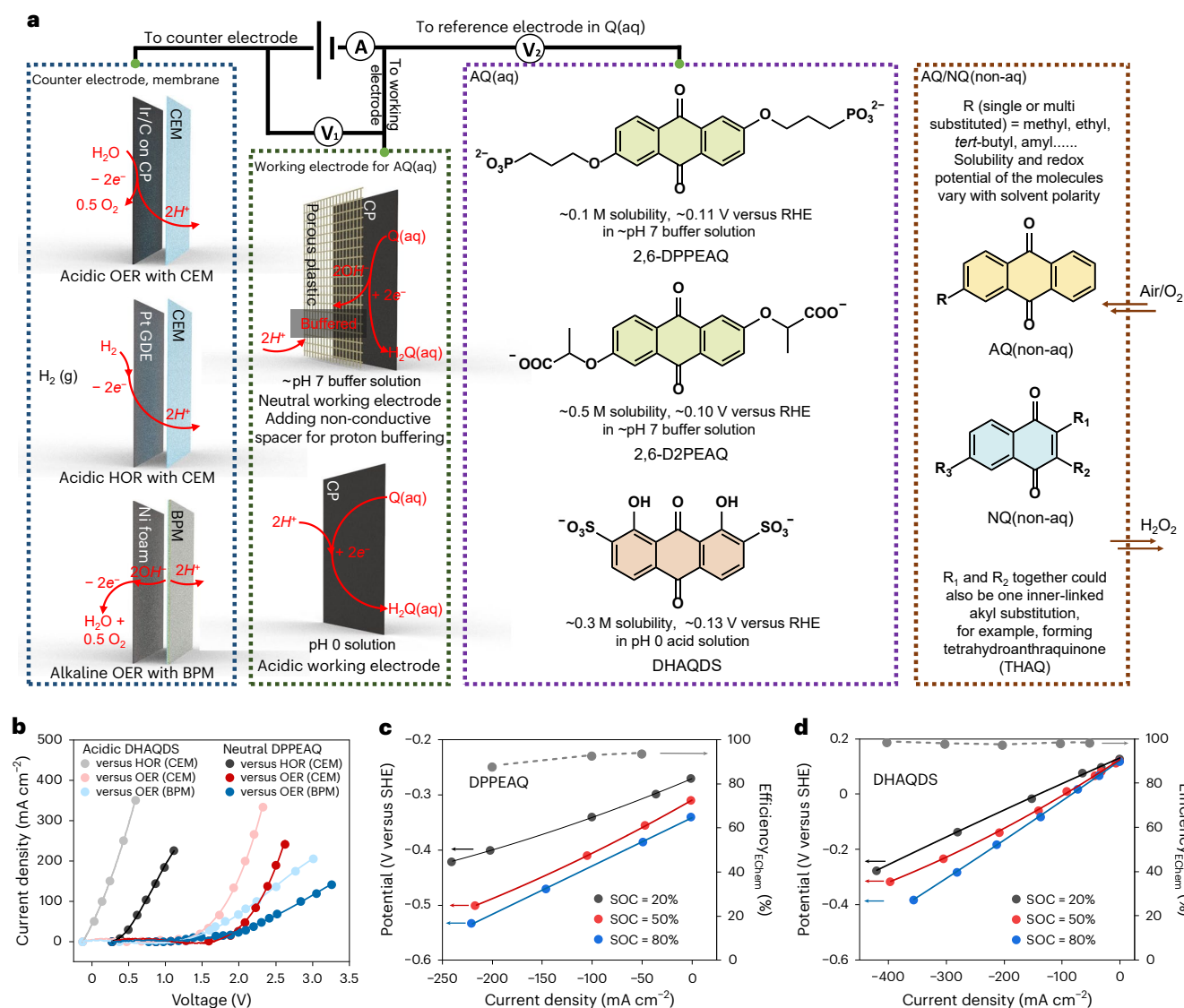


Fig. 4 | General e-AO process. **a**, Schematics of the e-AO process with flexible selection of counter electrode reactions, aqueous quinones and non-aqueous quinones. The circuit for electrochemical tests is drawn at the top. Cell structures, materials, molecules, reactions and charge carrier flows are labelled accordingly. **b**, Full-cell polarization performance of DPPEAQ (20% SOC) and

DHAQDS (20% SOC) against acidic HOR, acidic OER and alkaline OER. **c**, Polarization (half-cell, converted to versus SHE) and efficiency_{EChem} of DPPEAQ (pH 7). **d**, Polarization (half-cell, converted to versus SHE) and Coulombic efficiency_{EChem} of DHAQDS (pH 0).

as the ionic charge carriers into the cathode are protons. For example, we show that the hydrogen oxidation reaction (HOR) against a cation exchange membrane (CEM), acidic oxygen evolution reaction (OER) against a CEM or alkaline OER against a bipolar membrane (BPM) can be used on the anode side²¹. These reactions, molecular candidates and charge carriers are depicted in Fig. 4a. During our tests, the cell was monitored with two voltmeters, one recording the full-cell voltage (full-cell voltage V_1 , positive), another connecting the cathodic working electrode and a reference electrode inserted in aqueous AQ catholyte (referenced V_2). V_2 was converted to cathode voltage versus standard hydrogen electrode (SHE).

The full-cell polarization test was conducted using the catholytes of 0.1 mol l⁻¹ buffered DPPEAQ (pH 7, 20% SOC) or 0.25 mol l⁻¹ 1,8-dihydroxy-9,10-anthraquinone-2,7-disulfonic acid (DHAQDS, pH 0, 20% SOC) against acidic HOR, acidic OER and alkaline OER (Fig. 4b). V_2 was added to the full-cell voltage V_1 to gain insight into the properties of the counter reactions and cross-membrane voltage drop (Supplementary Fig. 22). Notably, the cells working with an acidic catholyte delivered better polarization performance than those

working with a pH-neutral catholyte, due to the elimination of the pH gradient and the reduction of ion polarization resistance across the membrane. For each kind of counter reaction, the HOR has the lowest energy cost but consumes hydrogen gas and, in our setup, utilizes a Pt/C gas diffusion electrode. The acidic OER with a CEM does not use hydrogen or a GDE, yet commonly requires noble metal catalysts, such as IrO₂. The alkaline OER with a BPM is a noble metal-free alternative²¹. The greater polarization resistance compared to the acidic OER is attributable to the higher resistance of the BPM, indicating a need for further development of more effective BPMs³³. Pure water can be used for zero-gap anode OER reactions to avoid long-term crossover of supporting acids or bases from the anolyte into the catholyte, without dramatically influencing cell polarization (Supplementary Fig. 23).

The hydrogenation rates of the aqueous molecules, equalling the product of current density, cell area and efficiency_{EChem}, can be determined by electrochemical characterization of the half-cell. We measured the polarization performance of DPPEAQ and DHAQDS half-cells under different SOC and the corresponding Coulombic efficiency at low SOC (Fig. 4c,d). The Coulombic efficiency drop for

DPPEAQ when the current density went higher than 100 mA cm^{-2} is caused by the pH gradient generated in the cell. Locally low pH on the near membrane side of the cathode under high current density boosts the hydrogen evolution (Supplementary Fig. 24). DHAQDS showed lower overpotential than DPPEAQ, delivering 400 mA cm^{-2} at -0.3 V versus SHE with 98.8% Coulombic efficiency. This is due to the uniform pH distribution in the acidic solution when using protons as charge carriers across the membrane. To show the practical feasibility of our method, we further tested the polarization of 2,2'-(9,10-dioxo-9,10-dihydroanthracene-2,6-diyl)bis(oxy)dipropionic acid (D2PEAQ), which is a considerably cheaper and synthetically accessible molecule, demonstrating higher solubility (0.5 mol l^{-1}) near-neutral pH³⁴. D2PEAQ also has a slightly more negative redox potential than DPPEAQ, resulting in a larger K_{eq} at the ANIPCET step (Supplementary Fig. 25).

The selection of aqueous/non-aqueous molecular pairs and non-aqueous solvents offered an opportunity for tuning the kinetics and thermodynamics during electrochemical hydrogenation and thus affecting the mass transport, ratio of reduction for non-aqueous molecules and efficiency under steady-state operation. For example, tetrahydroethylantraquinone (THAQ), although exhibiting slower AOR kinetics^{26,35}, is commonly used in the industrial AO process, having a redox core of a naphthoquinone, which generally has a higher redox potential and higher solubility than (ethyl)anthraquinone. The higher potential of the non-aqueous mediator relative to the aqueous mediator results in a higher K during ANIPCET (Supplementary Fig. 26). Flexibility of molecules and operating conditions are expected to lead to other applications and further optimization. Yet, without the formation of a suitable ANIPCET complex at the interface (such as a quinhydrone in the case of anthraquinone couples)²⁷, interfacial charge and proton transfers can have a higher energy barrier and slow kinetics^{36,37}. For instance, vanadium ($\text{V}^{2+} | \text{V}^{3+}$) in acid, although having a redox potential 500 mV lower than DPPEAQ, can hardly mediate ANIPCET under vigorous stirring, even for hours.

Techno-economic analysis shows the potential of commercializing e-AO using green electricity (Supplementary Fig. 27). The electricity cost dominates the levelized cost; therefore, cheap, green electricity is crucial for e-AO to become commercially attractive. For further optimization, an electrochemical cell with a uniform pH, a quinone pair that facilitates greater ANIPCET into the non-aqueous phase and a faster AOR with sequential mixer-settlers and sequential extraction should enable more efficient production, even with a variable power source.

Outlook

Achieving highly selective and efficient electrochemical synthesis in non-aqueous phases remains a substantial challenge due to the non-ideal proton sources and high resistance of these environments. However, non-aqueous liquid phases are often essential for specific reactions, offering advantages, such as optimal solvent polarity for reactions or automatic phase separation from aqueous systems for product separation. By contrast, aqueous-phase electrochemistry is typically more cost-effective and efficient due to lower ionic resistance, but often struggles with separating products from the aqueous electrolyte³⁸. We have demonstrated a solution to this dilemma by developing an electrified hydrogen peroxide production system.

Through the investigation of the mechanism and kinetic behaviour of ANIPCET, we electrified the conventional AO process using aqueous electrochemistry. The e-AO process, with its flexible choices in molecule, solvent and anodes, is capable of operating at high current densities with great Faradaic efficiency, and of yielding high-concentration, pure H_2O_2 solution. The process further improves the selectivity of the incumbent AO process and eliminates the use of hydrogen gas and metals for the hydrogenation side. With future chemical and molecular engineering regarding non-aqueous solvents and ANIPCET redox pairs, the e-AO process has the potential to be scaled up, further diminishing organic waste and leading to better performance. The e-AO process

can also facilitate the electrification of other applications that require peroxides, such as the synthesis of alkyl oxides, selective oxidation of chemicals and water treatment³⁹.

More broadly, this work electrifies non-aqueous synthesis by bridging non-aqueous chemistry and aqueous electrochemistry through interfacial chemical reactions. Unlike traditional phase-transfer catalysis, ANIPCET or other aqueous–non-aqueous interfacial coupled ion–electron transfer (ANICIET)⁴⁰ processes selectively transport only charge-neutral fragments (such as electron–proton pairs), rather than amphiphilic molecules, between phases. The formation of soft interfaces, across which highly selective and facile mass and energy transport occur, avoids the introduction of unwanted ions or phase-transfer catalysts in certain phases, diminishing the requirement for expensive purification or separation^{25,41}. With judicious selection of the chemical intermediates and phases that are selective to charge-neutral CIET and feature automatic phase separation, the kinetics of the ANICIET can be sufficiently rapid^{37,40} for scalable electrosynthesis with straightforward product separation^{11,42}.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41557-025-01940-7>.

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Methods

Materials and synthesis

DPPEAQ was purchased from TCI Chemicals²⁴. Nafion 212 as cation exchange membrane and Fumasep FBM as bipolar membrane were purchased and only soaked in deionized water before use. Ir/C, Ni foam, hydrophobic carbon paper and 5 mg cm⁻² Pt GDE were purchased from Fuel Cell Store. Potassium permanganate (KMnO₄), sodium oxalate (Na₂C₂O₄), dipotassium phosphate (K₂HPO₄), monopotassium phosphate (KH₂PO₄), sulfuric acid (H₂SO₄), potassium hydroxide (KOH), potassium ferricyanide (K₃Fe(CN)₆), potassium ferrocyanide (K₄Fe(CN)₆), EtAQ, 2,6-dihydroxyl-9,10-anthraquinone (DHAQ) and 1,8-dihydroxy-9,10-anthraquinone-2,7-disulfonic acid, sodium salt were purchased from Sigma-Aldrich. DHAQDS was prepared using a cation exchange resin column filled with Amberlyst 15(H) to exchange sodium ions for protons⁸. D2PEAQ was synthesized according to our previous report³⁴.

Electrode preparation

Electrodes for aqueous electrolyte tests were carbon papers (SGL 39AA) baked at 400 °C overnight. Carbon paper loaded with 1 mg cm⁻² Ir/C served for acidic OER. Typically, 10 mg of Ir/C catalysts, 0.97 ml of 2-propanol and 30 µl of Nafion binder solution were mixed to form a catalyst ink. The ink was sonicated for 30 min and then spray-coated onto carbon paper. Ni foam loaded with NiFe hydroxides was used for alkaline OER. The electrode was prepared by electrodeposition of NiCl₂·6H₂O (0.1 mol l⁻¹) and FeSO₄·7H₂O (0.05 mol l⁻¹) under -10 mA cm⁻² for 1 min, and was washed with deionized water.

H₂O₂ detection

The concentration of H₂O₂ was detected by a standard KMnO₄ titration process. KMnO₄ solution was prepared by dissolving 7.9 g KMnO₄ in 1 l of 1 N H₂SO₄ (about 0.05 mol l⁻¹ KMnO₄, 0.5 mol l⁻¹ H₂SO₄), and then was boiled and stored in brown reagent bottles in the dark. The exact concentration of KMnO₄ was calibrated by titrating with a standard Na₂C₂O₄ solution at 70 °C.

Analytical concentration and SOC detection

The analytical concentration of DPPEAQ and H₂DPPEAQ were measured by UV-Vis spectrophotometry (Agilent Cary 60 spectrometer). A solution without the target molecules was used as the 'blank', with the absorbance of the 'blank' automatically subtracted. We defined the analytical concentration (*c_A*) as the total concentration of DPPEAQ and H₂DPPEAQ, and the SOC as the concentration of H₂DPPEAQ divided by *c_A*. In each test, the electrolyte sample was collected and diluted to the detection range with 1 mol l⁻¹ KOH in a nitrogen-filled glovebox. The absorbance within the wavelength range of 200 nm to 800 nm was then recorded by UV-Vis spectrophotometry (Agilent Cary 60 spectrometer). The absorbances at wavelengths of 355 nm and 408 nm were ascribed to DPPEAQ (SOC = 0%) and H₂DPPEAQ (SOC = 100%), respectively. By varying *c_A*, we calibrated the concentration-absorbance curves. The concentration-absorbance curve of EtAQ was calibrated following the same procedures.

We obtained DPPEAQ solutions with a series of SOC values by mixing 0.05 mol l⁻¹ DPPEAQ and 0.05 mol l⁻¹ H₂DPPEAQ in varying proportions. The samples were diluted as stated. The absorbance at a wavelength of 355 nm (*A₃₅₅*) exhibited a strong linear relationship with the SOC.

Solubility test

The solubility of DPPEAQ and EtAQ was measured by adding each compound to the corresponding solution under sonication until no further solid could be dissolved. The saturated solution was then diluted, and its concentration was determined by UV-Vis spectrophotometry. The concentration was calculated according to a pre-calibrated absorbance-concentration curve. In addition, the solubility of DPPEAQ in

the non-aqueous phase and that of EtAQ in the aqueous phase were measured following the same procedures without the dilution step.

ANIPCET kinetics measurement

The kinetic rate constants involved in ANIPCET were measured using several non-aqueous phase polarities in a nitrogen glovebox. In a typical measurement, the aqueous phase consisted of a solution containing 0.05 mol l⁻¹ of DPPEAQ dissolved in 0.33 mol l⁻¹ phosphate buffer with a pH of about 7. The non-aqueous phase was 0.05 mol l⁻¹ EtAQ dissolved in a certain solvent mixture. Both the aqueous and non-aqueous phases were purged of dissolved O₂ with nitrogen before being transferred into the glovebox. DPPEAQ was first charged to a SOC of 100% under a CCCV protocol against buffered ferro/ferricyanide. The cutoff voltage for 20 mA cm⁻² constant current was set as 1.2 V, and the voltage was held until the current density dropped in magnitude to below 0.2 mA cm⁻². Extra phosphoric acid was added to adjust the final pH of the H₂DPPEAQ solution back to 7.3. Then, 15 ml of H₂DPPEAQ was transferred to a 50-ml centrifuge tube and 15 ml of EtAQ was added on the top of H₂DPPEAQ carefully, without mixing or splashing. The aqueous-non-aqueous interfacial area was 6.2 cm². In the meantime, two KNF diaphragm pumps consistently circulated the aqueous and non-aqueous phase, respectively, at 19.8 ml min⁻¹ (if not stated otherwise) while not disrupting the interface between them. A timer was started when the non-aqueous solution was added. The concentrations of the DPPEAQ and EtAQ solutions and the compositions of the non-aqueous solvents were changed as stated elsewhere. During each test, approximately 100 µl of the aqueous phase was intermittently collected and diluted 250 times to the detection range with 1 mol l⁻¹ oxygen-free KOH in a glovebox. The absorbance within the wavelength range of 200 nm to 800 nm was then recorded by UV-Vis. The absorbances of 0.05 mol l⁻¹ H₂DPPEAQ at 355 nm (*A_{355-red}*) and 0.05 mol l⁻¹ DPPEAQ at 355 nm (*A_{355-ox}*) were diluted in the same way and were measured before each test as the reference, to avoid the relative error of volume during sampling. The absorbance of the samples during the test at the wavelength of 355 nm is denoted as *A₃₅₅₋₁*. Subsequently, the solution in the cuvette was bubbled with air and shaken vigorously to ensure the H₂DPPEAQ was fully oxidized. Then, we conducted another UV-Vis measurement and recorded the absorbance at 355 nm, denoted as *A₃₅₅₋₂*. Using $\frac{A_{355-ox}}{A_{355-2}}$ to correct the volume of the initial sampling, we obtained the concentrations of H₂DPPEAQ and DPPEAQ during the test based on equations (8) and (9):

$$C_{DPPEAQ} = \frac{A_{355-1} \times \frac{A_{355-ox}}{A_{355-2}} - A_{355-red}}{A_{355-ox} - A_{355-red}} \times 0.05 \text{ mol l}^{-1} \quad (8)$$

$$C_{H_2DPPEAQ} = \frac{A_{355-ox} - A_{355-1} \times \frac{A_{355-ox}}{A_{355-2}}}{A_{355-ox} - A_{355-red}} \times 0.05 \text{ mol l}^{-1} \quad (9)$$

The kinetics test of ANIPCET under vigorous stirring was performed in a glovebox, using identical volumes and concentrations of aqueous and non-aqueous solutions. In a mixer, after 3 seconds (longer when necessary) of vigorous stirring and standing still until clear phase separation, 0.5 ml of the aqueous phase was taken out and diluted for UV-Vis. When no more transfer of proton-electron pairs between phases was observed, attainment of equilibrium was assumed. The SOC of the EtAQ was determined by sampling 200 µl of the non-aqueous solution, extracting the reduced form of EtAQ using 100 ml 1 mol l⁻¹ oxygen-free KOH and then measuring the absorbance of the obtained KOH solution with UV-Vis. The sampling, extraction and dilution for the determination of the SOC of EtAQ after ANIPCET were carried out three times (*n* = 3).

ANIPCET tests for various redox couples

The efficiency of ANIPCET and the spontaneity of ANIPCET with different combinations of aqueous anthraquinones and non-aqueous

quinones in various non-aqueous solvents were determined by measuring the extent of reaction between 0.05 mol l^{-1} aqueous H_2AQ and 0.05 mol l^{-1} non-aqueous quinones in the N_2 glovebox. First, 5 ml aqueous AQ solutions were charged to 100% SOC with CCCV. The recorded capacity was regarded as the overall capacity. Next, 5 ml non-aqueous quinone solution was added to the aqueous phase. After sufficient stirring, the emulsion was allowed to settle until it separated into two distinct phases. Recharging the separated aqueous solution using the same CCCV protocol allowed for the determination of the electron–proton transferred throughout the ANIPCET process. The extent of reaction was determined by dividing the transferred electron–proton by the total capacity. UV–Vis spectrophotometry was used to confirm the state of charge of aqueous AQs before and after ANIPCET.

Cycling of DPPEAQ

The cycling test of DPPEAQ was conducted by employing the CCCV protocol. A cell composed of 6.5 ml 0.1 mol l^{-1} phosphate-buffered DPPEAQ paired with 100 ml of 0.08 mol l^{-1} ferrocyanide + 0.02 mol l^{-1} ferricyanide. Excess capacity in the ferro/ferricyanide side ensured that DPPEAQ was the capacity-limiting side of the cell. The two half-cells were separated by a Nafion 212 membrane. Two layers of carbon paper were used in each half-cell. The flow rate of electrolytes was set to 60 ml min^{-1} . The cell was cycled at a constant current of 20 mA cm^{-2} , followed by a constant charging voltage of 1.2 V and a discharging voltage of 0.3 V until the current dropped to 0.2 mA cm^{-2} , accessing more than 95% of the theoretical capacity of the DPPEAQ side.

The cycling test for DPPEAQ + EtAQ was conducted using the same protocol. DPPEAQ initially underwent 32 consecutive cycles, followed by the addition of 3 ml of 0.1 mol l^{-1} EtAQ in 3Tol/2Dec onto the top of it. The solution was gently mixed by pumping during the cycling. Subsequently, an additional five cycles were conducted.

Performance test of H_2O_2 production

The batchwise e-AO was conducted to evaluate the overall efficiency. A portion of 10 ml of 0.1 mol l^{-1} buffered DPPEAQ and 50 ml of 0.5 mol l^{-1} EtAQ in 3Tol/2Dec was charged against the acidic OER. While charging, N_2 gas was continuously purged above the solution to prevent oxidation of H_2EtAQ by O_2 . As the SOC of EtAQ went higher, the colour of the non-aqueous phase turned dark green. After the SOC of EtAQ reached approximately 20% the charging was stopped, and the aqueous solution was removed from the non-aqueous phase. Next, the non-aqueous phase was exposed to air for oxidation. Deionized water in varying proportions to the non-aqueous phase was then added to extract the produced H_2O_2 . The extraction was carried out twice. The concentration of the generated H_2O_2 was titrated using KMnO_4 .

The continuous e-AO was conducted in a tandem reactor²¹. Specifically, 18 ml DPPEAQ (0.1 mol l^{-1} DPPEAQ in 0.33 mol l^{-1} phosphate buffer, pH 7) and 18 ml pure water were stored in the catholyte tank and the product tank, respectively, and continuously pumped at 50 ml min^{-1} through the corresponding mixers and separators. Then, 30 ml of 0.25 mol l^{-1} EtAQ phase was added and circulated on top of them. The flow rate was 20 ml min^{-1} for the non-aqueous phase. When a negative potential was applied to the cathode, DPPEAQ was hydrogenated and transported to Mixer 1 (Fig. 3a). In Mixer 1, the DPPEAQ phase mixed with the EtAQ phase under vigorous stirring, ensuring that ANIPCET was conducted thoroughly. Subsequently, the EtAQ phase carrying H_2EtAQ was transported to Mixer 2, mixing with air and the H_2O phase. The generated H_2O_2 accumulated in the product tank, and the oxidized EtAQ was circulated back to Mixer 1, moving on to the next reaction round. The mass flow of air was controlled to minimize oxygen residue in the non-aqueous phase to avoid the unwanted oxidation of H_2DPPEAQ . In total, two aqueous phases (a solution of DPPEAQ and one containing the H_2O_2 product) and one non-aqueous phase (EtAQ solution) were used and cycled individually.

During the continuous production of H_2O_2 , a current of 250 mA (9.33 mmol h^{-1}) was used, with a corresponding area of 5 cm^2 (equalling 50 mA cm^{-2} , $1.87 \text{ mmol cm}^{-2} \text{ h}^{-1}$). During the electrolysis, the cathode potential was recorded using a voltmeter connected to the cathode and a reference electrode inserted in the catholyte. Every 200 s, 0.5 ml of the aqueous phase was taken out from the H_2O_2 collection tank for H_2O_2 detection. Before and after the electrolysis, electrochemical impedance spectroscopy measurements were conducted with a 10 mV perturbation and with a frequency ranging from 1 to 100,000 Hz to determine the high-frequency area-specific resistance.

During the operation, the EChem step (an areal reaction) can be operated with a higher current by using a higher current density or a larger cell area, with the ANIPCET step reaching near equilibrium in less than 3 seconds. The maximum current that the setup can hold is limited by the relatively slow AOR (a volumetric reaction), which can be accelerated by increasing the temperature or oxygen partial pressure. Alternatively, a larger Mixer 2 with a larger volume of non-aqueous EtAQ can be used to match a higher current during EChem.

Polarization test

The full-cell polarization test was conducted with different combinations of anode reactions, membranes and cathode pH. The potential was swept from open circuit voltage at a scan rate of 100 mV s^{-1} using linear sweep voltammetry (LSV) controlled through the full-cell voltage V_1 , and the resulting current response was measured. During tests, the cell was monitored with two voltmeters, one recording the full-cell voltage (V_1) and another recording the cathode potential (V_2). The corresponding current was recorded.

The polarization performance of the cathode was evaluated using 0.1 mol l^{-1} buffered DPPEAQ, 0.25 mol l^{-1} buffered D2PEAQ and 0.25 mol l^{-1} DHAQDS in 0.5 mol l^{-1} H_2SO_4 . The cell was first charged to a certain SOC with respect to theoretical capacity. Then, LSV was conducted by sweeping the full-cell voltage V_1 at a scan rate of 100 mV s^{-1} , with half-cell voltage V_2 measured using a reference electrode inserted in the catholyte. The half-cell LSV and open circuit voltage results (converted to versus SHE) were determined based on V_2 and the corresponding current density.

Quantum chemistry computations

Detailed computational methods, which include all metadata and formula development, are described in the Supplementary Information. Generally, the geometry optimization of the quinhydrone analogue intermediate was conducted based on DFT, as performed using the Vienna ab initio simulation package. The calculation of the potential energy surface of ANIPCET was accomplished using the Gaussian 16 package.

Data availability

All data are available in the Article and Supplementary Information, and are available via figshare at <https://doi.org/10.6084/m9.figshare.29817470> (ref. 43).

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Acknowledgements

This research was supported by The Harvard Climate and Sustainability Translational Fund.

Author contributions

D.X. conceived the idea. D.X. and Y.W. designed and conducted hardware design, tests and electrochemical experiments. Z.Y. studied the hydrogenation and performance of non-aqueous anthraquinones

using hydrogen gas. Y.L. did the DFT calculation. R.Y.L. supervised the reaction mechanism study. M.J.A. supervised the project. D.X., Y.W. and M.J.A. drafted the manuscript. All authors edited the manuscript.

Competing interests

Harvard University has filed a patent application (PCT/US25/10441) based on this work with D.X., Y.W. and M.J.A. as inventors for ‘Electrochemically reducing an aqueous-soluble, non-aqueous-insoluble molecule in an aqueous electrochemical cell to provide ANIHAT to a molecule from which H₂O₂ is produced upon exposure to O₂’. M.J.A. is co-founder of Adiabatic Materials, a startup company that has licensed this intellectual property from Harvard. The remaining authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41557-025-01940-7>.

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Peer review information *Nature Chemistry* thanks Joaquim Henrique Teles and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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